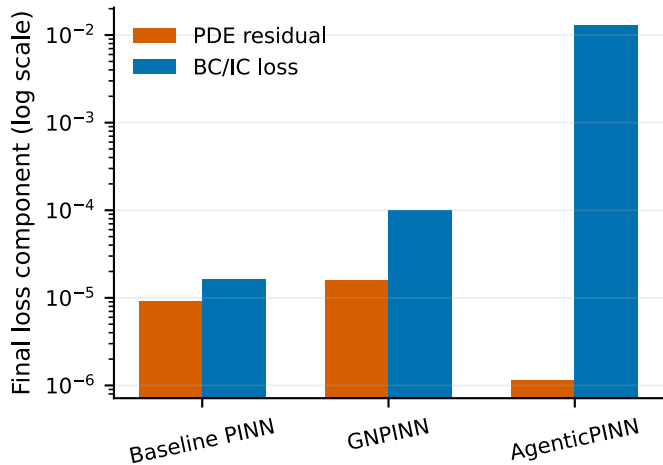


Component-level PINN findings and CFD surrogate validation

(a) Crystal-growth MLP trade-off



(b) Corrected CFD case-wise holdout

